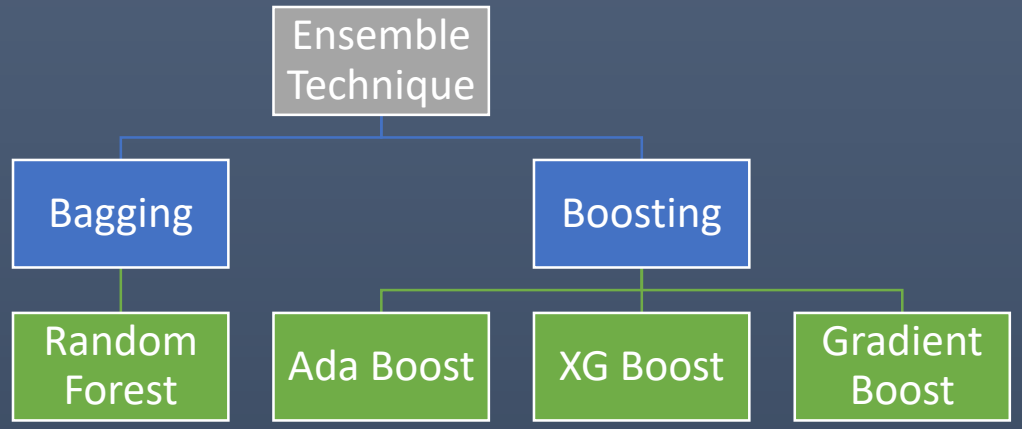
Ensemble techniques:

Ensemble techniques is **combining multiple models together to make one better prediction**.



Random Selection

Random selection means choosing data points from a dataset **randomly, without following a fixed order or preference**.

With Replacement vs Without Replacement

Let's use a simple dataset:

A, B, C, D, E

1. With Replacement

After selecting a data point, **put it back** before selecting the next one.

Without Replacement

After selecting a data point, **do not put it back**.

Example:

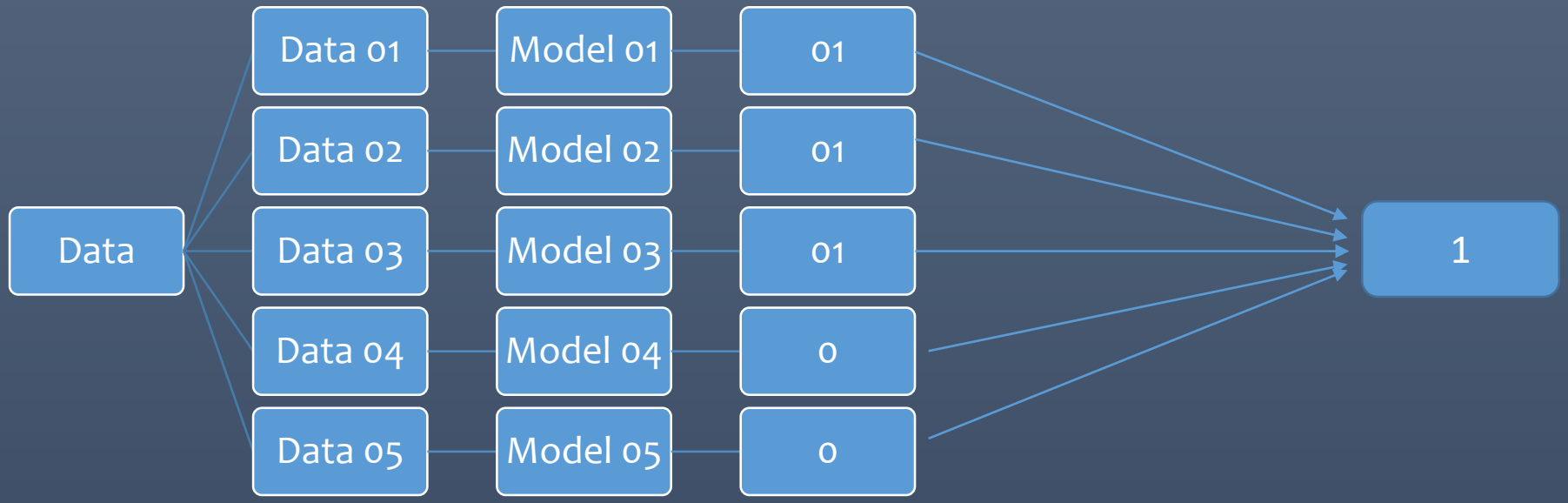
Bagging:

where **multiple models are trained on different**

- **randomly created samples of the training dataset**
- **with replacement**

and their predictions are **combined** to produce a more accurate and stable result.

1) Random Forest



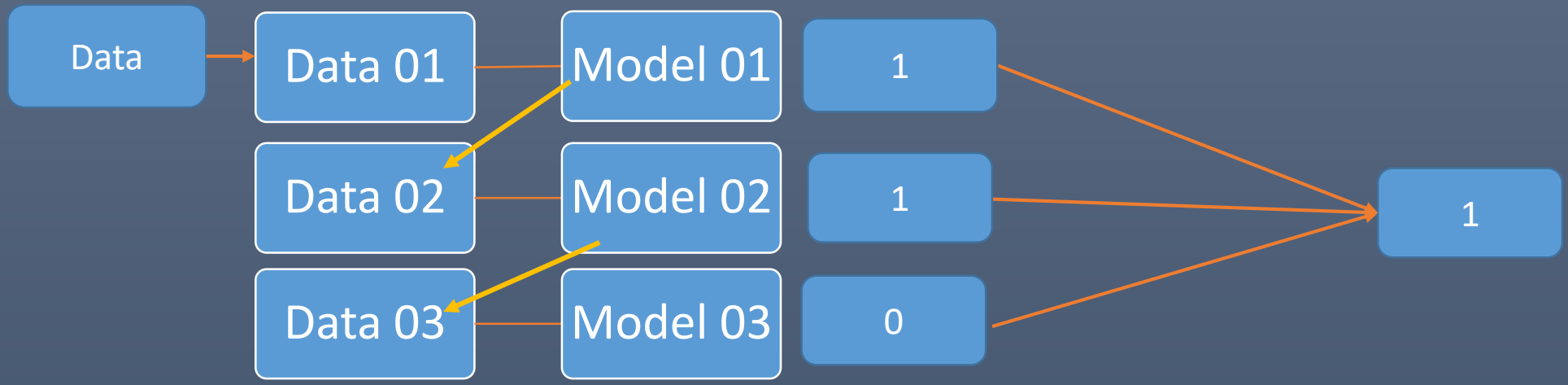
Combine their predictions.

1. **Classification:** majority voting
2. **Regression:** average of predictions

Boosting:

Boosting is an ensemble machine-learning technique where **models are trained one after another**, and each new model focuses more on the mistakes made by the previous models.

Student	Study Hours		Actual	Model 1
A	1	Fail	Fail ✓	
B	2	Fail	Pass ✗	
C	3	Pass	Pass ✓	
D	4	Pass	Fail ✗	
E	5	Pass	Pass ✓	



Boosting: -

- **Boosting** combines multiple **weak** or **base** models to create a stronger and more accurate predictive model.
- It works by **sequentially** train the weak models in such a way that each subsequent model focuses on the instances that were incorrectly predicted by the previous models.
- The final prediction is made by **aggregating** the predictions of all the weak models, typically using a weighted voting or averaging scheme.

Boosting algorithms, such as

1) **AdaBoost**
2) **XGBoost**
3) **Gradient Boosting**

have been widely used and have achieved **state-of-the-art** results in various machine-learning tasks.

AdaBoost — Simple Steps

AdaBoost (Adaptive Boosting) combines several **weak learners** (usually small decision trees) to create a strong classifier.

1. **Give equal weights** to all training data points.
2. **Train a weak learner** (e.g., a small decision tree).
3. **Check the errors** made by the learner.
4. **Increase the weights** of incorrectly classified points.
5. **Train the next weak learner** using the updated weights.
6. **Repeat** steps 3–6 for several learners.
7. **Combine all learners** using a weighted vote:
    1. Better-performing learners get **more importance**.
    2. Poorer-performing learners get **less importance**.

Classification
Pass / Fail

Model	Prediction	Weight
Model 1	Pass	0.4
Model 2	Fail	0.2
Model 3	Pass	0.6

Pass = 0.4 + 0.6 = 1.0
Fail = 0.2

Since **Pass has the higher total**, the final prediction is:
Classification → combine model votes/scores → choose the class with the highest combined score.

2. Regression

Regression predicts a **number.**

Temperature	Salary	House price
30°C	₹ 60,000	₹50 lakh

Model 1 correction = +₹5 lakh
Model 2 correction = −₹2 lakh
Model 3 correction = +₹3 lakh

₹50 lakh + ₹5 lakh − ₹2 lakh + ₹3 lakh
= ₹56 lakh

Regression → combine the numerical contributions of the models → get the final numerical prediction.